

390449

Course Name: MACHINE LEARNING

Course Outcome

- CO1- Apply the basic concepts of Machine Learning.
 CO2- Apply the concept of regression and resampling methods.
 CO3- Design the supervised and re-enforcement learning based solution.
 CO4- Apply the ensemble methods for improving classification algorithms.
 CO5- Identify the ways of feature extraction, reduction and selection.
 CO6- Design the applications of machine learning algorithms.

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University Roll No.

End Term Examination, Even Semester 2024-25

B.Tech. (CS/Biotech), 3rd Year, VI Semester

BCSE0105; Machine Learning

Time: 3 Hours

Maximum Marks: 50

Section – A

Attempt All Questions

4 X 5 = 20 Marks

No.	Detail of Question	Marks	CO	BL	KL
1	Building a churn prediction model with a highly imbalanced dataset. The team suggests using k-fold cross-validation for model evaluation. How would you set up cross-validation to handle class imbalance, and why might you prefer stratified k-fold cross-validation in this scenario to improve performance?	4	1	U	F
2	Given a dataset with continuous features and a binary target variable. You need to predict whether a customer will churn (1) or not (0) based on their features like tenure and monthly charges. Explain the key differences between linear regression and logistic regression in this context. Why would logistic regression be a better choice for this problem compared to linear regression? How would you interpret the coefficients in a logistic regression model compared to a linear regression model?	4	2	E	P
3	Explain the various feature selection methods (e.g., filter, wrapper, embedded) and how they help in selecting relevant features? Also, describe feature extraction method like PCA, focusing on how they	4	5	R	F

	transform the original data into a lower-dimensional space to enhance model efficiency and performance.				
4	Explain the purpose of Support Vector Machine in classification tasks. How does SVM determine the optimal hyperplane for separating data into different classes? Describe how SVM identifies the marginal line and explains the role of support vectors in maximizing the margin to ensure better generalization.	4	3	R	C
5	You are working as a data scientist for an e-commerce platform and need to classify customer reviews into positive and negative categories. Each review contains words like "great," "poor," and "disappointed." How would you apply the Naive Bayes algorithm to classify the reviews based on the frequency of these words? Discuss how you would handle feature independence and any potential limitations in real-world applications of Naive Bayes.	4	3	E	P

Section – B

Attempt All Questions

3 X 5 = 15 Marks

No.	Detail of Question	Marks	CO	BL	KL
6	Discuss the role of kernel functions in Support Vector Machines. Explain any three commonly used kernel functions along with their mathematical formulations.	3	3	U	C
7	You are working on a regression problem with many features, some of which might be irrelevant or highly correlated. To improve the model's performance and prevent overfitting, you decide to apply regularization. Explain the difference between Ridge (L2) and Lasso (L1) regularization. How does each regularization technique affect the coefficients of the model?	3	5	U	F

8	Explain the working of the Hierarchical Clustering algorithm. Differentiate between Agglomerative and Divisive approaches. Additionally, discuss the role of dendrogram in Hierarchical Clustering.	3	6	R	C
9	You are building an ensemble model for a classification problem. Compare Bagging and Boosting techniques in terms of model performance, bias-variance tradeoff, and how they combine multiple weak learners. When would you prefer boosting over bagging, and how does each method handle model overfitting?	3	4	R	C
10	A task with predicting customer satisfaction using a dataset with several categorical and numerical features. While a decision tree model performs adequately, it tends to overfit. How would Random Forest improve performance over a single decision tree in this scenario? Discuss how Random Forest reduces overfitting and enhances accuracy through ensemble learning.	3	6	An	C

Section – C

Attempt All Questions

5 X 3 = 15 Marks

No.	Detail of Question	Marks	CO	BL	KL
11	You are given the following 2D data points: (1,3), (4,7), (5,8), (10,12), (12,15), (14,18) Perform K-means clustering with K=3 clusters. What are the final centroids after convergence, and how are the points assigned to each cluster after the first iteration?	5	1	U	P
12	Given the following positive and negative data points: (3, 1), (3, -1), (6, 1), (6, -1) - Positive data points	5	1	E	P

	(1, 0), (0, 1), (0, -1), (-1, 0) - Negative data points Apply Support Vector Machine with a linear kernel to classify these points. 1. Determine the optimal hyperplane separating the two classes. 2. Calculate the margin between the hyperplane and the nearest points from each class (support vectors). 3. Identify the support vectors.																									
13	Given the following data points, apply the Agglomerative Hierarchical Clustering algorithm to construct the corresponding dendrogram. Use the minimum distance (Single Linkage) criterion to iteratively merge clusters, and update the proximity matrix at each step accordingly. <table><tr><th>Data Points</th><th>X</th><th>Y</th></tr><tr><td>A</td><td>4</td><td>3</td></tr><tr><td>B</td><td>1</td><td>4</td></tr><tr><td>C</td><td>2</td><td>1</td></tr><tr><td>D</td><td>3</td><td>8</td></tr><tr><td>E</td><td>6</td><td>9</td></tr><tr><td>F</td><td>5</td><td>1</td></tr></table>	Data Points	X	Y	A	4	3	B	1	4	C	2	1	D	3	8	E	6	9	F	5	1	5	1	E	P
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